Major Project -2
End Semester Report
For
“Multiple Facial Detection Using Resnet50”

Submitted by:

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1. Abstract

Face detection plays a crucial role in various real-world applications, where accuracy and efficiency play a crucial role. However existing methods often struggle to effectively handle complex scenes, diverse conditions, limiting their practical utility. In this study, we propose a human-centred approach to address these challenges by leveraging the Siamese Network alongside a ResNet50 model.Our approach revolves around understanding and accommodating the needs and challenges faced by humans in real-world scenarios. By using Siamese networks, which excel in learning highly discriminative features, and incorporating the ResNet50 backbone, renowned for its depth and performance in image recognition, we develop a robust framework for multiple face detection. The proposed model is trained to not only detect faces but also to understand their relationships and context within the scene. This enables it to accurately localise multiple faces, even in instances of occlusion, varying lighting conditions, and partial obstructions. By prioritising human-centric considerations, our approach aims to improve the overall user experience and applicability of face detection systems.This study emphasises the importance of considering human factors in the development of computer vision systems. By focusing on real-world challenges and user needs, we strive to create more reliable, robust, and user-friendly face detection solutions with broader applicability and impact.

2. Introduction

The system using facial recognition is an innovative project aimed at automating the tracking process. By leveraging advanced facial recognition technology, this system streamlines the entire process and improves overall efficiency. The primary purpose of this project is to provide a reliable and efficient solution for facial detection. By integrating facial recognition technology, the system aims to accurately identify faces without the need for traditional methods like roll calls or barcode scanning. This not only saves time but also reduces the chances of errors and increases data accuracy. Additionally, this technology-driven approach promotes a more engaging and interactive learning environment for students, enhancing their overall educational experience. The project's scope follows the development and implementation of a facial recognition-based system. The system will be designed to capture and analyse facial features to identify people. The ResNet50 framework deploys the ResNet50 algorithm as a structure. This architecture uses several components, like initial input layer, convolutional layers, distinctive residual blocks, pooling layers, fully connected layers, and an activation function. ResNet50 introduces a concept known as a “residual block” and these blocks consist of dual convolutional layers, which skip the connections[1].“Siamese network” is a type of neural network architecture that has been first introduced by Bromley et al. [2] for signature verification work purposes. They have trained two similar Siamese neural networks having shared the same parameters and weights, which gave an output of two feature vectors when two input signatures are fed to the network. The outcome of two signatures is two vectors. These vectors are compared based on some distance measure that has been used as a loss function during learning. Later, the Siamese network has been incorporated into other aspects in computer vision tasks including face verification [3], one-shot image recognition [4]. The crux of the Siamese neural network is to learn general feature representations with a distance (or similarity) metric calculated from the feature vectors from two similar inputs (images in our case). Siamese neural network is a class of network architectures that usually contains two identical networks. The two networks have the same number of layers and

configurations with the same parameters and shared weights. The parameter updating in one network is reflected across the other network since the configuration is shared. This framework has been successfully used for dimensionality reduction in weakly supervised metric learning and face verification in [5]. The selection of a Siamese network over a typical Convolutional network for face recognition has been made because of its low computational requirement for training.

3. Problem Identification

Despite advancements in face detection, accurately detecting multiple faces in various real-world scenarios remains challenging. Existing methods often falter in handling occlusion, diverse lighting conditions, and crowded environments, leading to diminished performance. To address these limitations, there's a growing demand for advanced face detection systems that harness the capabilities of deep learning, particularly the Siamese ResNet50 model. However, current research lacks a comprehensive exploration of such models tailored specifically for multiple face detection tasks, leaving a gap in practical solutions. The problem at hand is to develop and evaluate a novel face detection system based on the Siamese ResNet50 model, with a focus on reliably detecting multiple faces in complex scenarios. This system needs to overcome challenges like occlusion, lighting variations, and crowded scenes while being adaptable across different demographic groups. The overarching goal is to enhance the accuracy, reliability, and usability of multiple face detection technology in real-world applications, ultimately improving its effectiveness in serving human needs.

4. Objectives

Primary objectives:

1. Develop a face detection system based on the Siamese ResNet50 model. 2. Train the system to accurately detect multiple faces in various real-world scenarios.

Secondary objectives:

1.Evaluate the system's performance against existing methods.
2. Enhance the system's adaptability to different lighting conditions and crowded environments

5. Literature Survey

A reasonable number of research articles have been available in recent years on the domain of face recognition. The primary focus is to apply learning generalised face recognition to work across domains.

In the paper by Guo et al. [6], they use a 28-layer ResNet as the backbone, but with a channel-number multiplier of 0.5. This model has been implemented on a pre-trained model and trained on the Full face dataset. Accuracy of over 99% has been achieved. In the paper by Ling et al. [7], they have introduced an attention-based mechanism that uses standard convolutional neural networks (CNNs), such as ResNet-50, ResNet-101. Their work has enabled more discriminative power for deep face recognition.

In the paper by Song et al. [8], they propose to discard feature elements that occlusions have corrupted. They identify face occlusions using PDSN(Pairwise difference Siamese Network)

out all over the face. It is not necessarily restricted to the face portion from the nose. They used the Facescrub dataset with training data of 0.5 million images and gained an accuracy of 99.20%.

In the paper by Guo and Zhang [9], they study face recognition of imbalanced train image classes under different lightning, age pose, and variations. They choose a 34 layer standard residual network (ResNet-34) as their feature extractor. They introduced a concept called Classification vector-centred Cosine Similarity to train a better face feature extractor. A concept is introduced called underrepresented-classes promotion, which effectively addresses the data imbalance problem in one-shot learning. The recognition coverage rate has been increased from 25.65% to 94.89% at the precision of 99% for one-shot classes, while still keeping an overall accuracy of 99.8% for regular classes.

6. Existing System

Facial recognition systems, powered by cutting-edge algorithms and machine learning, analyse facial features for identification. Despite their benefits in security and personalization, they grapple with privacy concerns and biases. Globally, efforts persist to improve reliability, inclusivity, and ethical standards in these systems.

7. Proposed Design

The proposed facial recognition system design is a comprehensive framework comprising cutting-edge algorithms for accurate facial feature extraction and recognition. It incorporates a diverse and extensive database to ensure inclusivity and mitigate biases. Privacy is prioritised through encryption techniques and stringent access controls. User-friendly interfaces enable seamless integration, while continuous monitoring facilitates ongoing improvement.

8. Methodology

Our methodology for developing a robust face detection system using the Siamese ResNet50 model is designed to be both thorough and understandable. We start by gathering a diverse dataset, ensuring that the images are consistent in size and quality. Then, we use the ResNet50 model to extract important facial features from these images, training it to recognize and encode these features accurately.

ResNet50

The ResNet50 architecture uses several components, like initial input layer, convolutional layers, distinctive residual blocks, pooling layers, fully connected layers, and an activation function.

First phase of our project starts with an input image. Before using the ResNet-50, the image is preprocessed, such as resizing it to a fixed size (in our case 100x100 pixels), normalising pixel values, and converting into the RGB format. Convolutional Layers are used when the preprocessed image is passed through multiple convolutional layers in ResNet-50. Every convolutional layer uses a set of filters to circulate over the image and extract features.

After completion of the steps like convolutional layers and residual blocks, the image features

are processed through pooling layers, such as average pooling or max pooling. In

ResNet-50, the final pooling layer is also known as the global average pooling layer. This layer computes the value of each feature map across all dimensions, resulting in a fixed-size feature vector . The output of the final pooling layer is pushed into a 1D vector and passed through one or more fully connected layers. These layers further process the features and map them to a lower-dimensional space, often containing the desired number of dimensions for the embeddings. The output of the final fully connected layer represents the embeddings or feature vector for the input image.

Siamese network Model

Next, we create a Siamese network, which is a specialised neural network architecture that compares pairs of facial embeddings to determine if they belong to the same person or not. This network is trained using pairs of embeddings organised into three folders: one for anchor images, one for positive pairs (images of the same person), and one for negative pairs (images of different people).

During training, we optimise the Siamese network using contrastive or L1 loss functions, tweaking parameters to ensure it learns to effectively discriminate between faces. Once trained, we evaluate its performance using separate validation datasets, assessing metrics like accuracy and precision to gauge its effectiveness. Finally, we put the trained model to the test on new, unseen data to ensure it performs well in real-world scenarios. Throughout this process, we document our methods and findings meticulously, ensuring transparency and reproducibility.

In essence, our methodology is a step-by-step approach that aims to develop a reliable face detection system capable of accurately identifying faces in a variety of situations.

9. UML Diagram

1. Flow Chart:



Fig 9.1: Flowchart for the system

2. **Use Case Diagram:** It expresses what are the main functions our system provides.
   Use Case Diagram for our system :

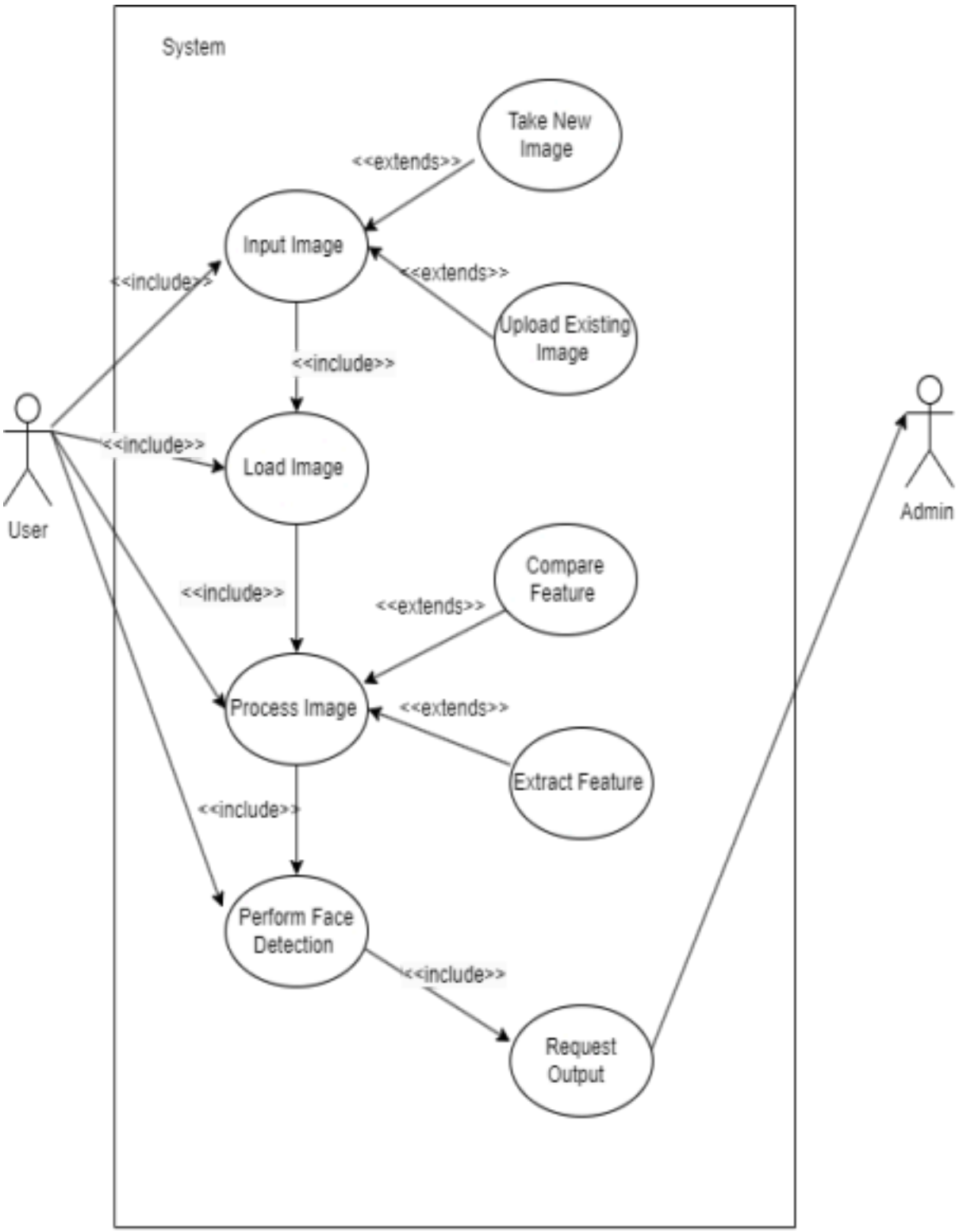


Fig 9.2: Use Case Diagram of the system

10. Results

The proposed model has been implemented with different configurations. The essential evaluation metrics for our project are : the configurations vary based on embedding vector

length, threshold, optimization algorithms, and the number of learning epochs, etc.

1. Precision :
   Training precision score : 0.9601572751998901
   Testing precision score : 0.980342447757721

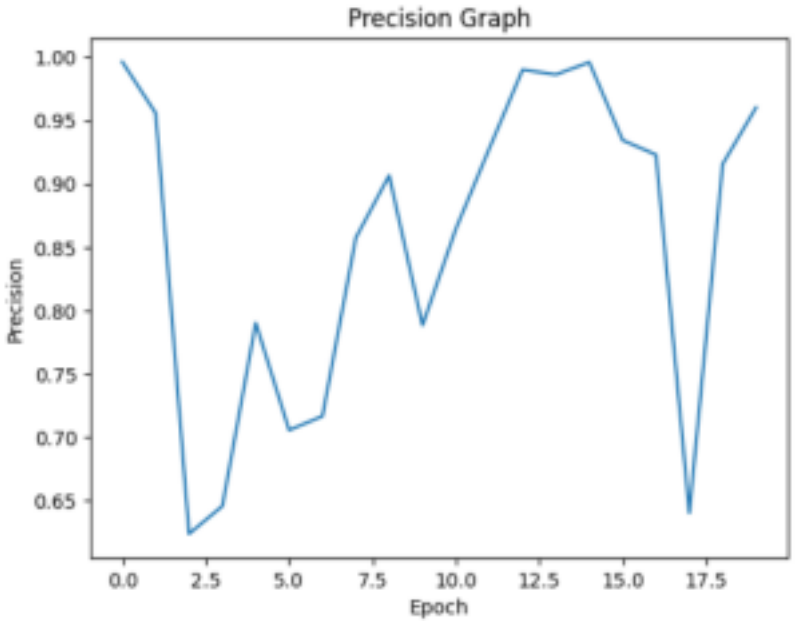


Fig 10.1

2. Recall:
   Training Recall Score : 0.9991816878318787
   Testing Recall Score : 0.9993535876274109

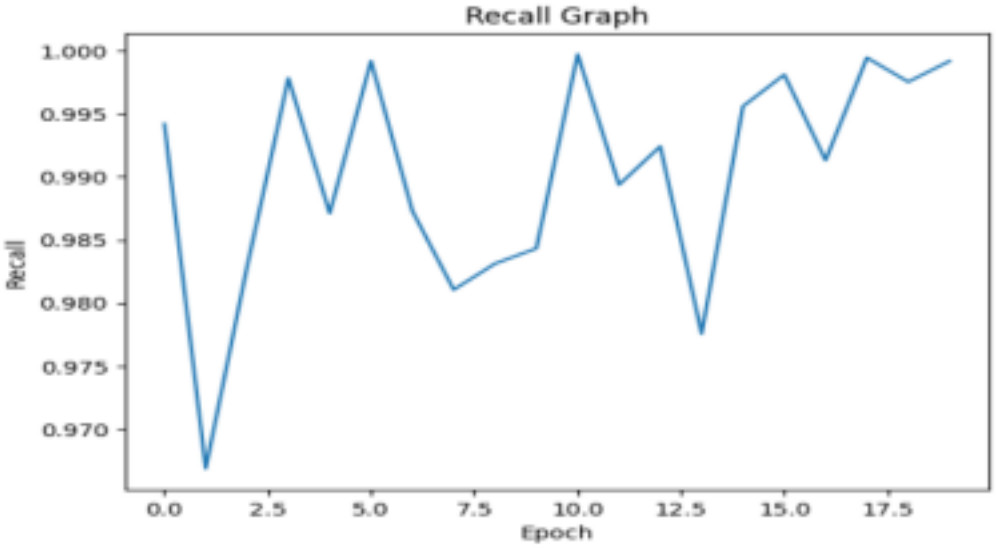


Fig 10.2

3. Loss:

Testing Loss score : 0.0004960999940522015

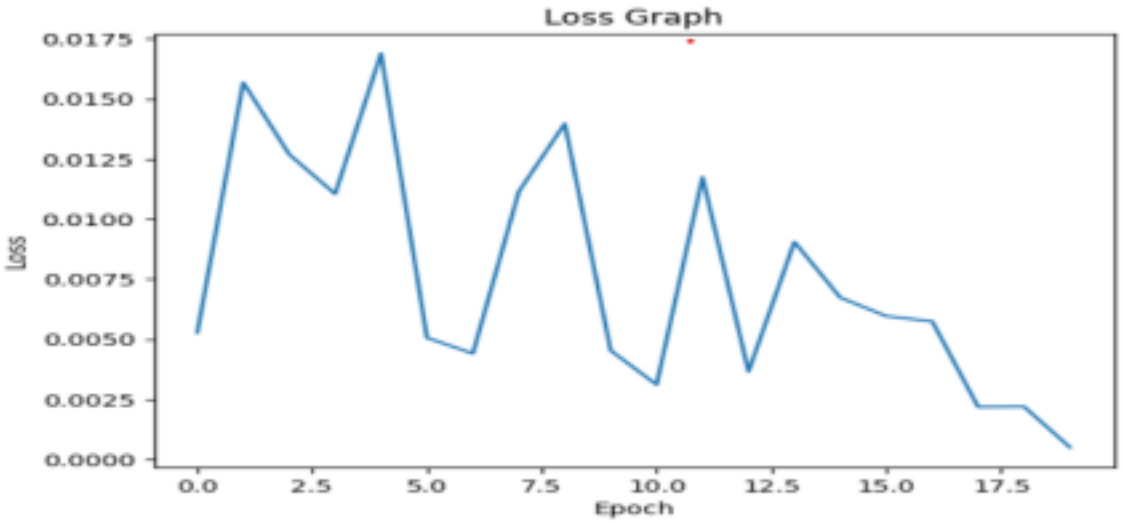
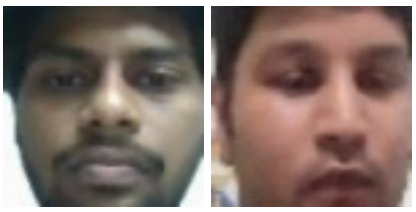


Fig 10.3

Image Testing In Real Time

People in image to be verified :

1. Saksham 2. Siddarth

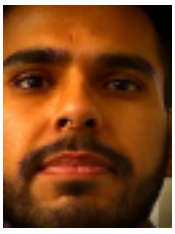


3. Tushar



Person In Unknown Image

1. Veer

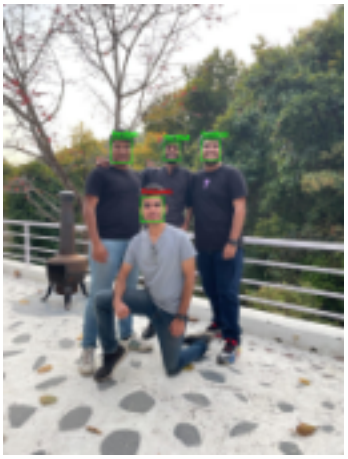


Facial Recognition In Final Images:

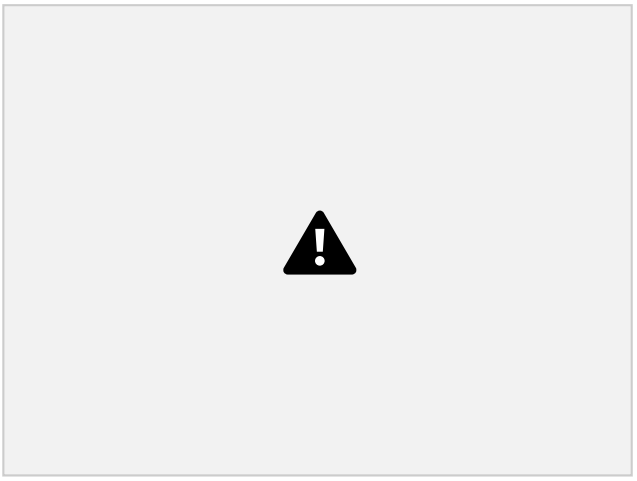












11. Conclusion

In summary, the proposed facial recognition system design offers a comprehensive solution to the challenges faced in accurately identifying individuals. Through advanced algorithms and a diverse database, the system ensures accuracy while minimising biases. Privacy concerns are addressed through encryption and strict access controls. The user-friendly interface simplifies integration, and ongoing monitoring allows for continual improvement. Overall, this system presents a promising approach to facial recognition, promising reliable performance across various applications.

12. References

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